

#HW 10

● Graded

Student

Aden Batool

Total Points

100 / 100 pts

Question 1

(no title)

30 / 30 pts

✓ - 0 pts Correct

Question 2

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 2 pts Mass flow rate = 0.53 kg/s

- 0 pts Click here to replace this description.

- 2 pts Temperature= -10 C

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt You have to add the two to get the total power -406 kW.

- 1 pt Power -406.2 kW (calculation is wrong.)

- 1 pt Power should be in negative.

- 1 pt Mass flowrate should be 2.2 kg/s.

- 6 pts incomplete

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt The answer will be positive.

- 6 pts incomplete

Question 5

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt The unit of temperature is degrees C.
- 1 pt The temperature will be around 575 degree C.
- 1 pt Work will be around 15-17 MW
- 6 pts incomplete

Question 6

(no title)

15 / 15 pts

✓ - 0 pts Correct

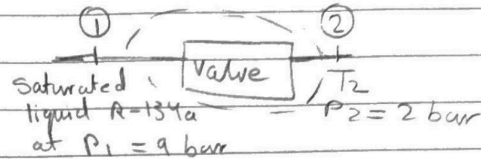
- 1 pt The second answer should be around 1327 s.
- 6 pts incomplete

Question assigned to the following page: [2](#)

HW 10

2. 7.18.67

a) find T_2 in $^{\circ}\text{C}$



$$M.B = \frac{dMc_v}{dt} = \sum \dot{m}_i - \sum \dot{m}_e$$

at steady state

$$\dot{m}_i = \dot{m}_e$$

$$-\dot{Q} = 0$$

$$-\Delta PE = \Delta KE = 0$$

- steady state

$$-\dot{m}_1 = \dot{m}_2$$

E.B 0 at s.s

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

$$0 = \dot{m} (h_1 - h_e)$$

$$h_1 = h_e$$

h_1 for saturated liquid R-134a at $P_1 = 9$ bar: 99.56 kJ/kg

T_2 at 99.56 kJ/kg and 2 bar:

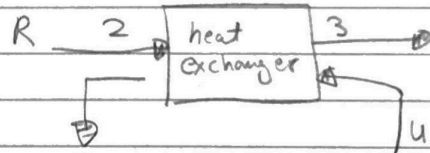
For 2 bar, T_{sat} is -10.09°C , $h_f = 36.84$ and $h_g = 241.30$

99.56 is between h_f and h_g so $T_2 = -10.09^{\circ}\text{C}$

b) mass flow rate of the refrigerant in kg/s

$$-\dot{m}_4 = \dot{m}_5 = \dot{m}_w$$

$$-\dot{m}_2 = \dot{m}_3 = \dot{m}_R$$



$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

giving heat losing heat

$$0 = \dot{m}_R (h_3 - h_2) - \dot{m}_w (h_4 - h_5)$$

$$-\dot{m}_w = 2 \text{ kg/s}$$

$$-h_2 = 99.56 \text{ kJ/kg}$$

$$-h_3 \text{ at } 2 \text{ bar and } 10^{\circ}\text{C} = 258.89 \text{ kJ/kg}$$

$$-h_4 \text{ at } 1 \text{ bar and } 25^{\circ}\text{C} = 104.89 \text{ kJ/kg}$$

$$-h_5 \text{ at } 1 \text{ bar and } 15^{\circ}\text{C} = 62.99 \text{ kJ/kg}$$

$$\dot{m}_R = \frac{\dot{m}_w (h_4 - h_5)}{(h_3 - h_2)}$$

$$\dot{m}_R = 2 \frac{(104.89 - 62.99)}{(258.89 - 99.56)}$$

$$\dot{m}_R = 0.526 \text{ kg/s}$$

$\approx h_f(T)$

Question assigned to the following page: [3](#)

3. Problem 7.18.11

a) Total $\dot{W}$ in kW?

- Steady state

- $\dot{Q} = 0$

- $\Delta PE = \Delta KE = 0$

- ideal gas

$$\dot{W}_{total} = \dot{W}_{cvA} + \dot{W}_{cvB}$$

Compressor A

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

$$\dot{W}_A = \dot{m}(h_i - h_e) = 0.6(300.19 - 607.02) = -184.098 \text{ kW}$$

h_i at 300 K: 300.19 kJ/kg

h_e at 600 K: 607.02 kJ/kg

Compressor B

$$\dot{W}_B = \dot{m}(h_i - h_e) = 0.6(451.8 - 821.95) = -222.09 \text{ kW}$$

h_i at 450 K: 451.8 kJ/kg

h_e at 800 K: 821.95 kJ/kg

$$\dot{W}_{total} = -222.09 + -184.098 = -406.2 \text{ kW}$$

b) $\dot{m}_{water}$ in kg/s

$$\dot{m}_2 = \dot{m}_3 = \dot{m}_{air}$$

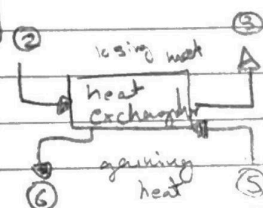
$$\dot{m}_5 = \dot{m}_6 = \dot{m}_{water}$$

$$0 = \dot{m}_{air}(h_3 - h_2) - \dot{m}_{water}(h_5 - h_6)$$

$$\dot{m}_{water} = \frac{\dot{m}_{air}(h_3 - h_2)}{(h_5 - h_6)} = \frac{0.6(451.8 - 607.02)}{(83.96 - 125.79)} = 2.23 \text{ kg/s}$$

h_5 at 20°C at 1 bar = $h_f(20^\circ\text{C}) \approx 83.96 \text{ kJ/kg}$

h_6 at 30°C at 1 bar = $h_f(30^\circ\text{C}) \approx 125.79 \text{ kJ/kg}$



Question assigned to the following page: [4](#)

4. Problem 7.18.72

Pump power required in MW?

$$\frac{dM_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e$$

at S.S $\dot{m}_i = \dot{m}_e$

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right)$$

$$0 = \dot{Q} - \dot{W} + \dot{m} (g z_i - g z_e)$$

$$\dot{W} = -0.6 \text{ MW} + \dot{m} 9.81 (0 - 20)$$

$$\dot{m} = \frac{\dot{V}}{v} = \frac{150 \text{ m}^3/\text{s}}{0.001 \text{ m}^3/\text{kg}} \quad \rho = \frac{1}{v} = \frac{1}{0.001 \text{ m}^3/\text{kg}} = 1000 \text{ kg/m}^3 = 0.001 \text{ m}^3/\text{kg}$$

$$\dot{m} = 150,000 \text{ kg/s}$$

$$\dot{W} = -600,000 + 150,000 (9.81) (-20)$$

$$= \frac{-30,300,000 \text{ W}}{10^6} = -30.3 \text{ MW} \rightarrow \boxed{30 \text{ MW}}$$

When the system generates on-peak electricity, the power will be less than the pump power because due to heat transfer lost and friction, the power generated will be less.

- Steady state

- electricity $\rightarrow \dot{W}$

- $\dot{V} = 150 \text{ m}^3/\text{s}$

- $\Delta z = 20 \text{ m}$

- $\Delta KE = 0$

- $\dot{Q} = -0.6 \text{ MW}$

- $\Delta P = 0$

- $\Delta T = 0$ $h_i = h_e$

Question assigned to the following page: [5](#)

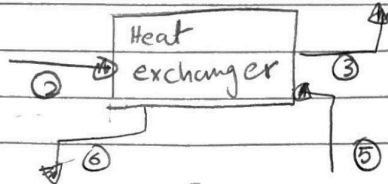
5 7.18.76

a) T_3 in K?

- Steady state

- $\dot{Q} = 0$

- $\Delta KE = \Delta PE = 0$



$$\dot{m}_2 = \dot{m}_3 = \dot{m}_w$$

$$\dot{m}_5 = \dot{m}_6 = \dot{m}_{air}$$

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right)$$

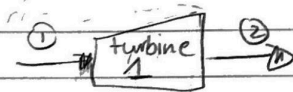
$$0 = (\dot{m}_w h_2 + \dot{m}_{air} h_5) - (\dot{m}_w h_3 + \dot{m}_{air} h_6)$$

$$\dot{m}_w h_3 + \dot{m}_{air} h_6 = \dot{m}_w h_2 + \dot{m}_{air} h_5$$

$$\dot{m}_w h_3 = \dot{m}_w h_2 + \dot{m}_{air} h_5 - \dot{m}_{air} h_6$$

$$\dot{m}_w h_3 = \dot{m}_{air} (h_5 - h_6) + \dot{m}_w h_2$$

$$h_3 = \frac{\dot{m}_{air} (h_5 - h_6)}{\dot{m}_w} + h_2$$



$$\dot{m}_1 = \dot{m}_2 = \dot{m}_w$$

$$\dot{W} = \dot{m}_w (h_1 - h_2)$$

$$10,000 = \dot{m}_w (3690.1 - 3263.9)$$

$$\dot{m}_w = 23.46 \text{ kg/s}$$

$$h_1 \text{ at } 600^\circ\text{C and } 20 \text{ bar} = 3690.1 \text{ kJ/kg}$$

$$h_2 \text{ at } 400^\circ\text{C and } 1 \text{ bar} = 3263.9 \text{ kJ/kg}$$

$$h_5 \text{ at } 1500 \text{ K} = 1635.97 \text{ kJ/kg}$$

$$h_6 \text{ at } 1200 \text{ K} = 1277.79 \text{ kJ/kg}$$

$$h_2 = 3263.9 \text{ kJ/kg}$$

$$\dot{m}_{air} = \frac{500 \text{ kg}}{\text{min}} \cdot \frac{1 \text{ min}}{60} = 25 \text{ kg/s}$$

$$h_3 = \frac{25(1635.97 - 1277.79)}{23.46} + 3263.9 = 3645.59 \text{ kJ/kg}$$

$$T_3 \text{ at } h_3 \text{ and } p_3 = \boxed{576^\circ\text{C}}$$

$$\frac{600 - T_3}{600 - 540} = \frac{3697.9 - 3645.59}{3697.9 - 3565.6}$$

Questions assigned to the following page: [5](#) and [6](#)

b) $\dot{W}_{t2} = ?$



$$\dot{W}_{t2} = \dot{m}w(h_3 - h_4)$$

$$h_3 = 3645.59 \text{ kJ/kg}$$

$$h_4 \text{ at } 240^\circ\text{C and 1 bar} = 2955.5 \text{ kJ/kg}$$

$$\begin{aligned} \dot{W}_{t2} &= 23.96 (3645.59 - 2955.5) \\ &= \boxed{16,190 \text{ kW}} \end{aligned}$$

6. Problem 7.18.87

a) the time required for the onset of evaporation is?

$$m = \frac{V}{v} = \frac{0.5}{1.0018} = 0.499 \text{ kg}$$

$$v \text{ at } 20^\circ\text{C} = 1.0018 \text{ m}^3/\text{kg}$$

$$Q = m c_p (T_2 - T_1) \quad c_p = 4.179 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$Q = (0.499)(4.179)(100 - 20)$$

$$Q = \frac{166.86 \text{ kJ}}{0.85 \text{ kW}} = \boxed{196 \text{ s}}$$

b) Total t for evaporation? once it starts

$$1 \text{ bar} \rightarrow h_{fg} = 2258 \text{ kJ/kg}$$

$$Q = m \cdot h_{fg} = 0.499 \text{ kg} \cdot 2258 \text{ kJ/kg} = \frac{1,126.742 \text{ kJ}}{0.85} =$$

$$\boxed{1,326 \text{ s}}$$